

Jit Malla

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Kolkata, India

Research Objective

Computer Science graduate and current M.Sc. student at the Indian Association for the Cultivation of Science (IACS) with strong interests in **computer vision**, **machine learning**, and **scene graph-based visual reasoning**. My research focuses on bridging visual perception and symbolic reasoning through deep learning, graph-based representations, and large language models (LLMs). I am particularly interested in understanding the **robustness, reliability, and limitations of multimodal AI systems**, and aim to pursue doctoral research in **Computer Science** with an emphasis on multimodal learning, interpretable AI, and intelligent systems with real-world impact.

Research Experience

M.Sc. Research Project — IACS Kolkata

2024 – Present

Scene Graph Generation and Spatial Reasoning using Vision-Language Models

- Developed a pipeline for **scene graph generation** from laboratory images using LLMs under two settings: image-only and image with bounding box annotations.
- Conducted a systematic study demonstrating that **bounding box annotations significantly improve object identification and spatial reasoning**, with measurable gains in recall and F1-score.
- Generated structured **scene graphs (objects and spatial relations)** and designed post-processing methods for improved **geometric consistency and relational accuracy**.
- Leveraged **LLMs for spatial reasoning and question answering** over scene graphs, incorporating an agent-based verification mechanism to improve logical consistency.
- Investigating challenges related to **model reliability, error propagation, and hallucination** in multimodal pipelines, with ongoing work toward **real-time and layered 3D scene graph representations**.

B.Sc. Project — Ramakrishna Mission Residential College, Narendrapur

2023 – 2024

A Wavelet-Based Approach for Authenticating Medical Images and Extracting Patient Information

Guide: Bibek Ranjan Ghosh, Assistant Professor

- Developed a secure medical image authentication framework using **wavelet transforms** for embedding and extracting patient metadata.
- Ensured preservation of **diagnostic image quality** while maintaining data integrity.
- Evaluated robustness against **compression, noise, and common image processing distortions**.

Technical Skills

Programming Languages: Python, C, C++, Java

Machine Learning & Computer Vision: Design and implementation of Convolutional Neural Networks (CNNs), Feature Extraction, Latent Space Visualization using t-SNE, Transformer-based Models

Graph-Based Vision & Reasoning: Scene Graph Generation, Object-Attribute Relationship Modeling, Spatial Relationship Inference, Graph-based Knowledge Representation

Tools & Frameworks: PyTorch, NumPy, OpenCV, Visual Studio Code, Linux/Unix

Documentation & Visualization: LaTeX, Graph and Scene Graph Visualization using PyDot

Relevant Coursework

Computer Science: Data Structures and Algorithms, Object-Oriented Programming (Java), DBMS, Operating Systems, Theory of Computation, Digital Image Processing, Algorithm Analysis, Software Engineering, Computer Networks, Artificial Intelligence & Data Science, Discrete Mathematics, Computer Architecture, Digital Electronics, Numerical Analysis, Machine Learning, Artificial Intelligence

Mathematics: Linear Algebra, Ordinary Differential Equation

Statistics: Descriptive Statistics (I & II), Probability Theory (I & II), Sampling Methods, Statistical Inference

Additional Information

- Qualified **GATE 2026 (Computer Science and Information Technology)**.
- Actively preparing for national-level eligibility examinations (NET) with the goal of pursuing doctoral research.
- Research interests include computer vision, graph-based representations, and vision-language-reasoning frameworks, with hands-on experience in scene graph-based spatial reasoning.
- Experienced in working within research-driven and collaborative academic environments, including iterative experimentation, presentation, and discussion of research ideas.